ARCHITECTURE & METHODS · VERSION 2.0

NeoStay
How it works

A complete, plain-language guide to the NeoStay platform: the data it stands on, the algorithms that forecast a newborn's hospital stay, the method that turns those forecasts into safe nurse staffing, and everything a hospital needs to run it.

Waseem Altaf, MD

Project and Clinical Lead · Neonatologist

Clinical direction, outcome interpretation, data stewardship, domain validation.

Abdul Razak, PhD

Technical Lead, Lead Developer and Applied AI/Data Science Lead

Architecture, algorithms, full-stack implementation, validation, deployment.

DOCUMENT
Architecture & methods reference

AUDIENCE
Clinical leads, nursing management, hospital IT

APPLICATION
NeoStay 2.0 · sign-in required

STATUS
Decision-support prototype — not a medical device

00 Contents

Everything in this document was read back from the source code before it was written down.

1 · What NeoStay does 3

2 · Words used in this document 4

3 · System architecture 5

4 · The data foundation 7

5 · Algorithm 1 — outcome prediction 8

6 · Algorithm 2 — nursing workload for one infant 10

7 · Algorithm 3 — the acuity classifier 12

8 · Algorithm 4 — unit staffing and the 24-hour roster 14

9 · Population analytics 3

10 · Access control and security 17

11 · Running NeoStay in a hospital 19

12 · How correctness is checked 20

13 · Adapting NeoStay to any hospital 21

14 · Future steps 23

15 · Appendix — files, endpoints, settings 24

How to read this document. Every section opens with a plain-language explanation for readers who do not write software, followed by the exact rule the program applies. Worked examples use real output from the application, reproduced by running the code itself. Where a number appears, it can be reproduced by running NeoStay with the inputs shown.

4
ALGORITHMS DESCRIBED

8
FIGURES AND DIAGRAMS

105
AUTOMATED TESTS PASSING

01 What NeoStay does

The problem, the answer, and who benefits.

A neonatal intensive care unit (NICU) cares for newborn babies who are very small or very ill. Two questions dominate every admission: **how long will this baby stay?** and **how many nurses — and which nurses — will this baby need?** NeoStay answers both, using ten years of the unit's own recorded outcomes and a published acuity tool, and presents the answers in a web application a clinician can use in under a minute.

For neonatologists

An expected length of stay with an honest range, the chance of going home, being transferred or dying, and how confident that estimate is — matched to babies of the same birth weight and gestational age.

For charge nurses

A live census that converts each baby's clinical findings into a nurse-to-patient ratio, then adds up the whole unit into nurses needed per shift, the right mix of experience, and a 24-hour roster of the nurses actually available.

For managers and educators

Population analytics across birth-weight bands and gestational weeks, and a shared reference for the acuity tool the unit uses.

The five screens

TABLE 1 — WHAT EACH PAGE OF THE APPLICATION DOES

Page	Question it answers	What it shows
Home	What is in the data?	Headline figures for the whole cohort and a chart of stay by birth weight.
Predict	What should we expect for this baby?	Length of stay with range, discharge outcome, survival, confidence, and the nursing the stay will need.
Scheduling	How many nurses tonight, and who?	Census built from acuity findings, nurses per shift, skill mix, 24-hour roster, uncovered cots.
Acuity tool	What are the rules?	The full acuity tool and nursing levels of care, exactly as transcribed from the source document.
Analytics	What does the population look like?	Stay and discharge outcomes across every weight band and gestational week.

What NeoStay is not. It is a decision-support prototype for planning and teaching. It is not a medical device, it has not been clinically validated, and it never replaces clinical judgment. It holds aggregated statistics only: no patient-level records exist anywhere in the system, and nothing typed into the tools is sent to a server.

02 Words used in this document

Plain meanings first. Every later section uses these terms exactly.

TABLE 2 — GLOSSARY

Term	Plain meaning
NICU	Neonatal intensive care unit: the ward for newborn babies who need intensive care.
VLBW	Very low birth weight: born under about 1,500 grams (3.3 lb). NeoStay's data covers these babies.
Gestational age	How many weeks the pregnancy lasted. Full term is about 40 weeks.
Length of stay	Days from admission to leaving the unit.
Disposition	How the stay ended: discharged home, transferred to another hospital, or died.
Median	The middle value. Half of similar babies stayed less, half stayed more.
Interquartile range (Q1–Q3)	The middle half of babies. A realistic range, not a best and worst case.
Acuity	How sick a baby is, and therefore how much nursing attention is needed.
Nurse:patient ratio	How many babies one nurse cares for. 1:1 means one nurse for one baby; 1:3 means one nurse for three.
NHPPD	Nursing hours per patient day: total nursing hours divided by days of stay.
Census	The list of babies currently admitted.
Skill mix	How many nurses of each experience level a shift needs.
Levels of care (N1–N4)	Four categories of nursery, from a newborn nursery (N1) to the most intensive NICU care (N4).
Benner levels (L1–L4)	A standard nursing career ladder: Novice, Competent, Proficient, Expert.

One convention throughout. **N1–N4** always describes the *baby's* level of care. **L1–L4** always describes the *nurse's* experience level. Section 7 explains how one is translated into the other.

03 System architecture

Three deployments, one codebase, two engines that must agree.

NeoStay is a web application whose calculations run **inside the user's own browser**. Nothing a clinician types is transmitted. The server's job is narrower: it checks who is signed in and hands out the reference data. That design keeps the application fast, private, and simple to host — either on a hospital server or on a public static host.

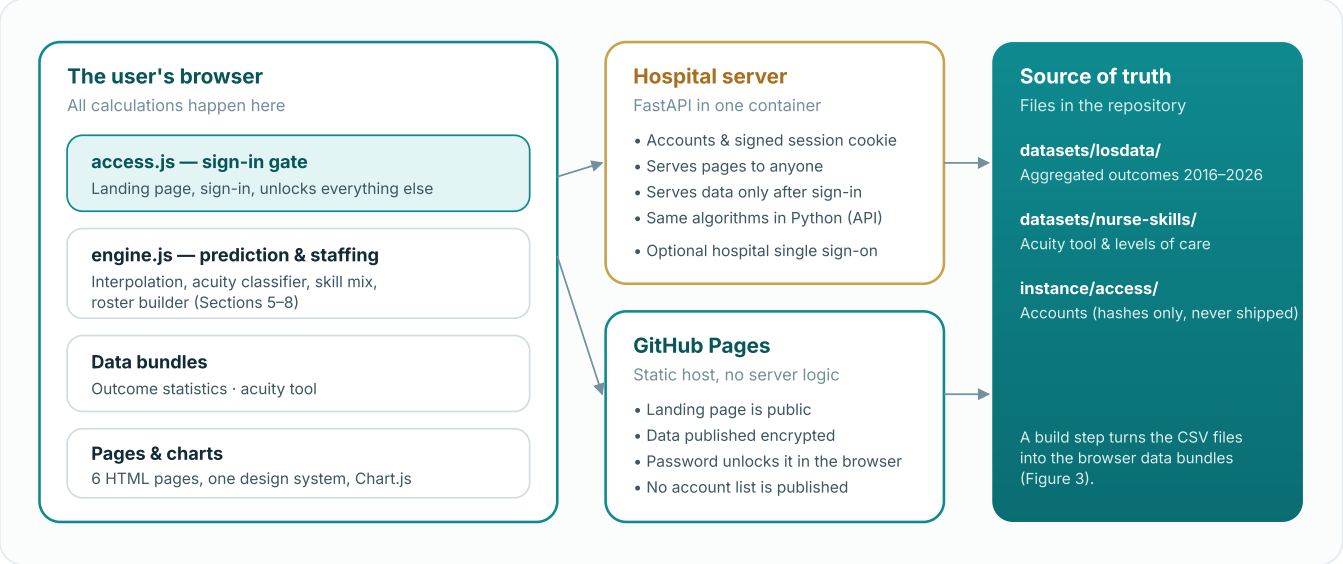


Figure 1 — Architecture overview. The browser holds the whole application. A hospital server and the public GitHub Pages copy are two ways to deliver it; both require sign-in, and both draw on the same files in the repository.

Two engines, kept identical

The same algorithms exist twice: in `frontend/engine.js` for the browser, and in `backend/app/` for the Python API. A test runs the browser engine in a JavaScript runtime and compares its answers with Python's, value by value, so the two can never drift apart (Section 12).

TABLE 3 — MAIN COMPONENTS AND WHERE THEY LIVE

Component	File	Responsibility
Sign-in gate	`frontend/access.js`	Landing page, sign-in form, loads data and page scripts only after sign-in.
Encrypted-data reader	`frontend/sealed.js`	Opens the encrypted data published on GitHub Pages, in the browser.
Browser engine	`frontend/engine.js`	Prediction, acuity classification, staffing, skill mix, rostering.
Shared layout	`frontend/components.js`	Header, navigation, footer, tooltips, sign-out.
Outcome data reader	`backend/app/data_loader.py`	Parses the aggregated CSV tables into bins and statistics.
Prediction	`backend/app/predictor.py`	Interpolation, blending, disposition probabilities, confidence.
Nursing model	`backend/app/nursing.py`	Severity, care phases, ratios, skill mix, roster construction.
Acuity classifier	`backend/app/acuity_tool.py`	Parses the acuity tool CSVs; classifies an infant from its findings.
Accounts and sign-in	`backend/app/accounts.py`, `auth.py`	Password hashing, session cookies, rate limiting.
Application	`backend/app/main.py`	Routes, access rules, security headers.
Build steps	`scripts/build_frontend_data.py`, `build_pages_site.py`	Data bundles for the browser; the published (encrypted) site.

04 The data foundation

What NeoStay knows, where it comes from, and how it reaches the browser.

NeoStay stands on two bodies of data. Neither contains a patient record. The first is a decade of the centre's own aggregated outcomes; the second is the acuity tool the unit classifies babies with. Nothing in the application is randomly generated: every figure it shows is computed from these two sources by rules set out in this document.

TABLE 4 — THE TWO DATA SOURCES

Source	Contents	Used by
datasets/losdata/ 3 CSV tables	For each of 11 birth-weight bands and 18 gestational-age groups: how many babies (N), the median stay, and the quartiles Q1 and Q3. A second table splits those by discharge outcome.	Prediction (Section 5), Analytics, Home page.
datasets/nurse-skills/ 3 CSV tables + scanned PDF	The Abrazo Arrowhead Neonatal Acuity Tool (11 body systems × 4 ratio columns) and the Nursing Levels of Care (45 general criteria, 19 hyperbilirubinemia criteria).	Acuity classifier (Section 7), Scheduling, Acuity tool page.

The scanned document is never read by the program. The acuity tool arrived as a three-page scan. Its tables were transcribed into CSV files, and only those CSV files are parsed. The handwritten annotations on the scan are deliberately ignored.

How the raw tables are shaped

The outcome tables are "wide": the first row names each band, the second row repeats four sub-columns beneath it (N, Median, Q1, Q3), and each later row is one measure, such as *Total Length Of Stay*. The loader reads that shape into one statistic per measure per band, and attaches a numeric midpoint to every band so that values can be interpolated between them.

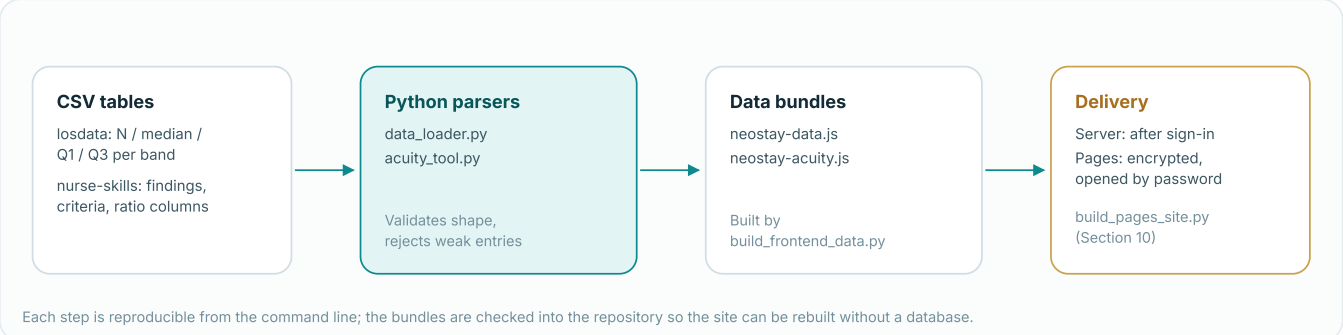


Figure 2 — From source tables to the browser. No database is involved at any point: the data is small enough to be prepared once and read directly.

The bands

Predictions are made against 11 birth-weight bands, from *under 501 g* to *over 1400 g*, and 18 gestational-age groups, from *under 22 weeks* to *over 37 weeks*. Each band carries a midpoint (for example 750 g for the 701–800 g band) which anchors the interpolation in Section 5.

05 Algorithm 1 — outcome prediction

From two numbers known at birth to a stay, an outcome and a confidence.

In plain words. NeoStay looks up what happened to babies of a similar birth weight, and to babies of a similar gestational age, then blends the two pictures. Because a 720 g baby sits between two recorded weight bands, the figures either side are mixed in proportion, so every gram and every week changes the answer slightly. Nothing is modelled or guessed: the numbers come from the unit's own counts.

The five steps

01 **Locate the bands.** The birth weight falls in one weight band (upper bound inclusive), the gestational age in one age group (upper bound exclusive).

02 **Interpolate.** For the median, Q1 and Q3, take the two nearest band midpoints and mix them in proportion to the distance. Values beyond the first or last midpoint are held at that end rather than extrapolated.

03 **Blend.** Average the weight-based estimate and the age-based estimate.

04 **Discharge outcome.** Take the recorded counts of Home, Transfer and Died for that weight band and convert them to percentages. Survival is 100% minus the died percentage.

05 **Confidence.** Report how many babies stand behind the answer, graded by sample size.

```
Interpolation between two band midpoints x₀ and x₁ (values y₀, y₁):

    y = y₀ + (value - x₀) / (x₁ - x₀) × (y₁ - y₀)

Blend of the two views:   LOS = mean(LOS by weight, LOS by gestational age)
Survival:                 survival % = 100 - died %
```

TABLE 5 — CONFIDENCE GRADES (BY NUMBER OF SIMILAR BABIES, N)

Grade	N	How to read it
high	50 or more	A solid basis for planning conversations.
moderate	20 – 49	Usable, but treat the range as wide.
low	5 – 19	A rough sketch only.
very low	under 5	Indicative; do not plan on it.

Worked example — a baby of 720 g born at 25 weeks

The inputs fall in the **701–800 g** band and the **25 Weeks** group. These are the figures NeoStay returns, reproduced by running the prediction code:

Quantity	Median	Q1	Q3
Length of stay by birth weight	103.6 days	83.8	128.0
Length of stay by gestational age	102.0 days	86.0	122.0
Blended forecast	**102.8 days**	**84.9**	**125.0**
If discharged home	105.8 days	94.2	122.6
If transferred	79.6 days	32.3	122.7
If the baby dies	6.7 days	2.2	19.1

Discharge outcome	Home	Transfer	Died	Survival	N	Confidence
Recorded for this weight band	68.8%	17.2%	14.1%	85.9%	64	high

Read as: babies like this one typically stayed about **103 days**, with the middle half between **85 and 125 days**. Roughly **seven in ten** went home. The shorter stay in the "died" row reflects deaths that occurred early in the admission — the application shows this rather than hiding it.

Honest limits. These are group averages, not a prediction for an individual baby, and they describe one centre's history between 2016 and 2026. A baby with a condition the data does not capture may behave very differently.

06 Algorithm 2 — nursing workload for one infant

Turning a forecast stay into nurse-hours, shifts and the experience required.

In plain words. A baby is not equally demanding throughout its stay. The first part is intensive, with one nurse at the cot; later the baby shares a nurse with one other, then with two others as it feeds and grows. NeoStay estimates how sick the baby is, splits the expected stay into those three phases, and adds up the nursing hours.

Step 1 — a severity score between 0 and 1

Two things drive it: how small the baby is and how early it was born. The admission condition and the breathing support then raise it.

```
weight part      = clamp01((1400 - birth weight in g) / 1000)
age part         = clamp01((34 - gestational age in weeks) / 12)
severity (S)     = clamp01(0.55 × weight part + 0.45 × age part + modifiers)
```

TABLE 6 — CONDITION MODIFIER

Condition on admission	Added
Stable	0.00
Guarded	+0.12
Serious	+0.25
Critical	+0.40

TABLE 7 — BREATHING SUPPORT MODIFIER

Respiratory support	Added
Room air	0.00
Nasal cannula	+0.05
CPAP	+0.15
Ventilator	+0.30

Step 2 — split the stay into three phases

```
intensive share     = 0.15 + 0.45 × S          → one nurse per baby   (1:1)
intermediate share  = 0.30 + 0.10 × (1 - S)    → one nurse per two    (1:2)
convalescent share  = whatever remains         → one nurse per three  (1:3)
(the three shares are normalised so they add to 1)
```

Step 3 — convert to hours, shifts and a minimum nurse level

```
phase days    = phase share × expected length of stay
nurse-hours   = phase days × 24 hours × nurses per baby
NHPPD         = total nurse-hours ÷ length of stay
12-hour shifts = total nurse-hours ÷ 12
```

The minimum experience level follows the acuity band, then rises for invasive support: severity 0.35 or more requires a Competent nurse (L2), 0.60 or more a Proficient nurse (L3); a ventilated or critically ill baby requires

at least L3; a baby that is both critically ill *and* ventilated requires an Expert (L4). Within a stay, the intensive phase carries the full requirement, the intermediate phase L2, and the convalescent phase L1.

Worked example — the same 720 g baby, serious on admission, on CPAP

Severity reaches its maximum of **1.00** (the small size and early birth already score high, and serious + CPAP add 0.40). With the forecast stay of 102.8 days:

Phase	Days	Nurses per baby	Nurse-hours	Minimum nurse
Intensive care (1:1)	61.7	1.00	1,480.3	L3 Proficient
Intermediate care (1:2)	30.8	0.50	370.1	L2 Competent
Convalescent care (1:3)	10.3	0.333	82.2	L1 Novice
Whole stay	**102.8**	—	**1,932.6**	**L3 at admission**

That is **18.8 nursing hours per patient day** and about **161 twelve-hour shifts** of nursing for this one baby.

Where the numbers come from. The ratios follow AAP and AWHONN acuity-based staffing standards. The phase shares, the severity weights and the experience thresholds are *modelling assumptions* for planning discussion, chosen to be easy to change — not values from a validated dataset. Section 13 shows where a hospital changes them.

07 Algorithm 3 — the acuity classifier

The unit's own tool, applied consistently to every cot.

In plain words. Instead of estimating how sick a baby is from weight and age, the charge nurse can record what is actually happening at the cot: what breathing support is running, what lines are in, which medicines are infusing. The acuity tool turns those findings into a nurse:patient ratio. A second table, the Nursing Levels of Care, decides which level of nursery the baby belongs to. NeoStay applies both, for every baby, the same way every time.

What was transcribed

11
BODY SYSTEMS

108
SELECTABLE FINDINGS

64
LEVEL-OF-CARE CRITERIA

Findings cover weight and age, respiratory, cardiovascular, A/B/Ds, nutrition, neuro, hyperbilirubinemia, transfusions, infection, lines, and haematology/other. The 64 criteria split into 45 general criteria and 19 hyperbilirubinemia criteria.

TABLE 8 — THE FOUR RATIO COLUMNS, AND WHAT NEOSTAY PLANS WITH

Column in the tool	Printed ratio	Used for planning	Counts as	Minimum nurse	Assessment frequency
Intensive care	1:2 or 1:1	1:2, or 1:1 when a footnoted criterion applies	0.50 / 1.00	L3 / L4	every 1–2 hours
Intermediate care	1:2–3	1:2	0.50	L2	every 2–3 hours
Continuing care	1:3	1:3	0.333	L1	every 3–4 hours
Continuing care (lower)	1:3–4	1:3	0.333	L1	every 3–4 hours

Where the printed tool gives a range, NeoStay plans with the more protective end. The single exception is intensive care, where the tool's own footnote defines which babies need 1:1 — the hemodynamically unstable

baby needing interventions every 15 minutes, therapeutic hypothermia, exchange transfusion, and bedside surgical procedures. Those findings are marked in the application with a 1:1 tag.

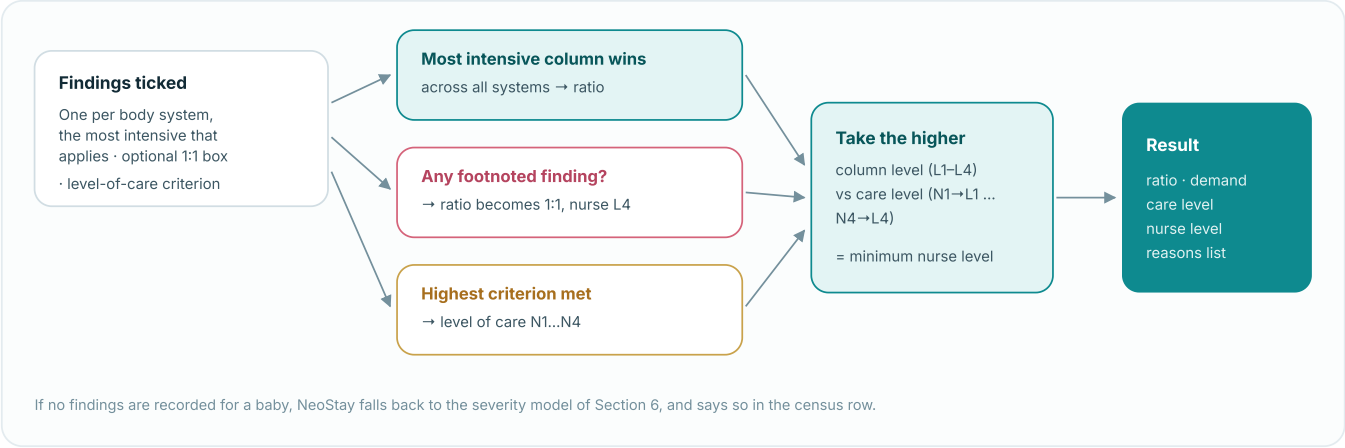


Figure 3 — How one baby is classified. The ratio comes from the most intensive finding anywhere on the tool; the nurse level is the higher of the column's requirement and the level of care.

Worked example — a ventilated baby on parenteral nutrition with a PICC line

Ticked on the tool	Column
Respiratory: "HFOV, Ventilator, NIPPV, CPAP/BCPAP, High flow or Vapotherm >4 liters"	Intensive care
Nutrition: "TPN/Lipids and/or IV fluids"	Intensive care
Lines: "Infusing IV, PICC, UVC/UAC"	Intermediate care
Level of care: "supplemental respiratory support ≥ 40%, NIPPV or mechanical ventilation or HFNC >1 L/min"	N3

Result: ratio **1:2** (intensive care, no footnoted 1:1 criterion), counts as **0.50** of a nurse, level of care **N3**, minimum nurse L3 Proficient, assessment **every 1–2 hours**. The application lists the three deciding findings when the ratio is hovered, so the number can always be traced back to a tick.

Had "Therapeutic Hypothermia/HIE" been ticked instead, the footnote would apply: ratio **1:1**, counting as a whole nurse, minimum L4 Expert.

08 Algorithm 4 — unit staffing and the 24-hour roster

From a census of cots to a defensible shift plan.

> **In plain words.** Each baby needs a fraction of a nurse. Add those fractions together and you have the bedside nurses the shift needs. But a number alone can hide an unsafe roster: three novices cannot replace three experts. So NeoStay also works out how experienced the team must be, then lays the nurses the unit actually has across the day and shows which cots, if any, are left without a qualified nurse.

Step 1 — demand

Every baby contributes its ratio: a 1:1 baby counts 1.00, a 1:2 baby 0.50, a 1:3 baby 0.333. The sum, rounded up, is the number of bedside nurses. One charge nurse is added per shift.

Step 2 — skill mix

Requirements accumulate downward: a nurse may always cover a baby below their level, never above it. NeoStay adds the demand at L4, then L4+L3, then L4+L3+L2, rounding each total up, and converts those running totals into a headcount per level. Two safety rules then apply: novices are capped at **30%** of the bedside team, with the surplus promoted to Competent, and every novice on shift needs a Proficient or Expert preceptor. Finally the roster's *effective capacity* is computed by weighting each nurse by experience, so a numerically adequate but junior shift is visible.

TABLE 9 — NURSE LEVELS, AS NEOSTAY MODELS THEM

Level	Stage	Experience	Counts as	Typical assignment
L1	Novice / Advanced Beginner	0–1 year	0.75	Convalescent babies; intermediate care with a preceptor.
L2	Competent	1–3 years	0.90	Intermediate and special care; stable intensive with support.
L3	Proficient	3–5 years	1.00	Intensive 1:1 assignments, including ventilated babies.
L4	Expert	5+ years	1.10	Most unstable babies, charge nurse, precepting.

Step 3 — the 24-hour roster

The nurses available across the whole day are dealt out shift by shift, most senior first and rotating, so no shift collects all the experts. Within a shift the most senior nurse becomes charge and carries no cot. The sickest babies are placed first, each into the *least senior qualified* nurse with room, keeping experts free for the babies who need them. No nurse exceeds one full assignment. If a baby cannot be placed, the charge nurse takes it as a last resort — reported as a warning — and anything still unplaced is listed, with the extra nurses that would close the gap.

Worked example — the sample unit built into the Scheduling page (8 babies, two 12-hour shifts, 20 nurses available)

#	Weight	Age	Ratio	Care level	Acuity band	Counts as	Minimum nurse
1	620 g	24 wk	**1:1**	N4	intensive	1.000	L4 Expert
2	740 g	25 wk	1:2	N3	intensive	0.500	L3
3	880 g	27 wk	1:2	N3	intensive	0.500	L3
4	980 g	28 wk	1:2	N2	intermediate	0.500	L2
5	1150 g	29 wk	1:2	N2	intermediate	0.500	L2
6	1300 g	30 wk	1:3	N2	convalescent	0.333	L2
7	1420 g	31 wk	1:3	N2	convalescent	0.333	L2
8	1650 g	33 wk	1:3	—	convalescent	0.333	L1

4.00
TOTAL DEMAND (NURSES)

5
NURSES PER SHIFT (4 BEDSIDE + CHARGE)

10
DAILY NURSE-SHIFTS

Level	Demand needing this level	Recommended on shift	Effective capacity
L4 Expert	1.00	1	1.10
L3 Proficient	1.00	1	1.00
L2 Competent	1.67	2	1.80
L1 Novice	0.33	0	0.00
Total	**4.00**	**4 bedside**	**3.90**

With 20 nurses in hand (4 · 6 · 6 · 4 across L1–L4), the roster covers every baby on both shifts: 10 nurses per shift, **48.0 of 120 rostered nurse-hours committed**, average bedside load 0.44 of a full assignment. The application states the verdict in words — here, "fully covered" — rather than leaving the comparison to the eye.

Reading the effective capacity. 3.90 against a demand of 4.00 means this particular mix is marginally junior for the work, even though the headcount matches. Adding one Expert instead of one Competent would close it. This is exactly the signal the number exists to give.

09 Population analytics

What a decade of recorded outcomes looks like, band by band.

In plain words. The Analytics page does not model anything. It draws the recorded figures exactly as the centre reported them, so that a reader can see the gradient the forecast rests on: smaller and younger babies stay longer and die more often. Nothing on this page is simulated, and no individual baby appears anywhere.

TABLE 10 — THE THREE CHARTS AND WHAT EACH ONE READS

Chart	Reads	Shows
Length of stay by birth weight	*Total Length Of Stay* for the 11 weight bands	A bar spanning Q1 to Q3, with the median as a line across the tops.
Length of stay by gestational age	The same measure for the 18 gestational-age groups	The same bar-and-median form, so the two gradients can be compared directly.
Disposition mix by birth weight	The by-disposition tables: home, transfer, died	Each band as percentages of its own admissions, stacked to 100%.

Every value is read from the same bundle the forecast uses, so a chart and a prediction can never disagree. Across the whole cohort of 708 infants with a recorded total stay, the median is 60 days with an interquartile range of 40 to 95 days.

TABLE 11 — THE GRADIENT, IN THE RECORDED FIGURES (SELECTED BANDS)

Birth-weight band	N	Median stay	Q1–Q3	Home	Died
< 501 g	38	43.5 d	10–149	38%	51%
701–800 g	64	100.0 d	77–125	69%	14%
901–1000 g	63	73.0 d	57–94	87%	5%
1201–1300 g	63	52.0 d	36–65	86%	10%
> 1400 g	105	42.0 d	25–54	85%	8%

Why the smallest band has the shortest median. At under 501 g, half of the admissions did not survive, and a death often ends the stay early. The median of 43.5 days therefore sits below the next band's 119.5 days, while the quartiles (10 to 149 days) show the two very different courses behind it. The page shows the disposition mix beside the stay for exactly this reason, and the Predict page separates the two in the same way (Section 5).

10 Access control and security

Registered users only — enforced differently on a server and on a static host.

> **In plain words.** Every page begins at a sign-in screen. On a hospital server the server checks the password and refuses to hand over any data until it is satisfied. On a public static host no server exists to do the checking, so the data is published encrypted and the password unlocks it inside the browser. Both use one account list.

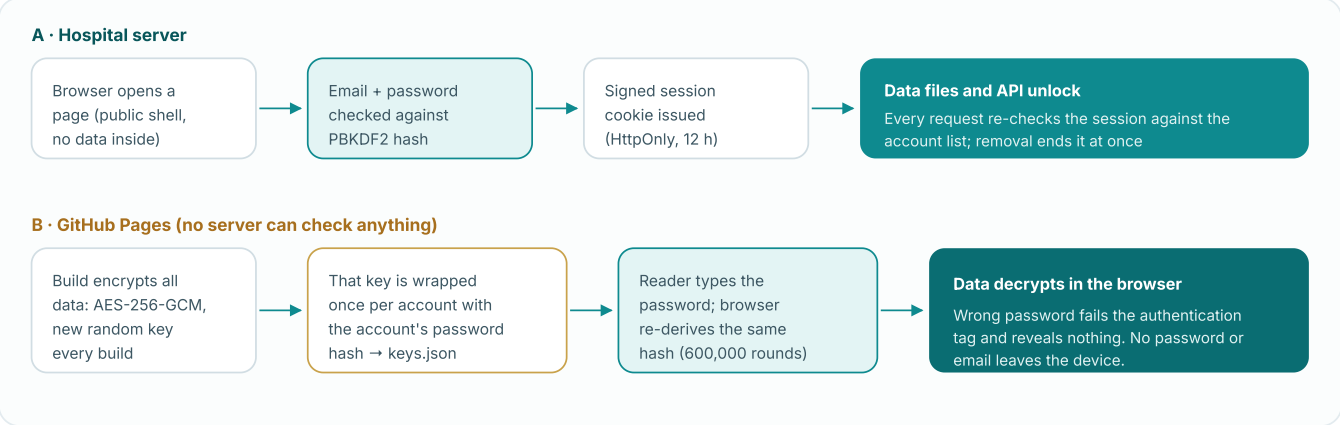


Figure 4 — Two ways to enforce one account list. The server path is stronger; the published path is what makes a static host usable at all. Section 10 states the limits of each.

TABLE 12 — SECURITY PARAMETERS, AS IMPLEMENTED

Control	Setting
Password storage	PBKDF2-HMAC-SHA256, 600,000 iterations, random 16-byte salt per account, 32-byte hash
Minimum password length	12 characters; the email address itself is rejected
Session	Signed token (HMAC-SHA256) in an `HttpOnly`, `SameSite=Lax` cookie, `Secure` in production, 12 hours by default
Session revocation	Every request re-checks the account and a fingerprint of its password: removing an account or changing its password ends its sessions immediately
Brute-force limits	5 failed sign-ins per account and 20 per client in 15 minutes, then a wait
Cross-site sign-in	Refused; an unknown email and a wrong password get the same answer, in the same time
Published data	gzip, then AES-256-GCM with a fresh 256-bit key per build; key wrapped per account; published files carry no email, name or password hash
Production start-up	Refuses to run without sign-in, or with a session secret under 32 characters, or with a wildcard host
Browser policy	Content-Security-Policy on every page, `X-Frame-Options: DENY`, HSTS in production, no caching of data or API responses

Two limits to state plainly. First, anyone can download the published encrypted files and try passwords offline, slowed only by the 600,000 hashing rounds — so accounts used on the public copy need long, unique passwords. Second, removing an account protects only copies published afterwards. A hospital server has neither limitation.

11 Running NeoStay in a hospital

One container, one address, accounts managed from the command line.

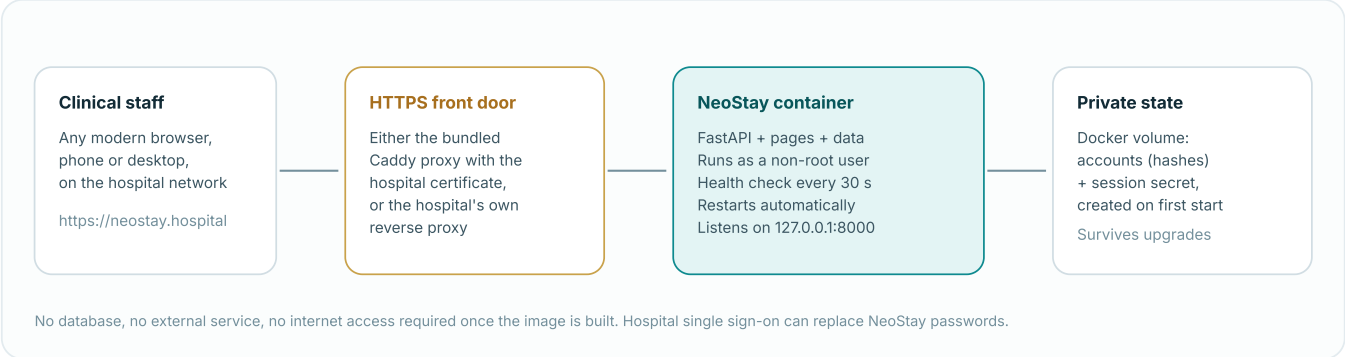


Figure 5 — Deployment topology. Everything a hospital runs is one container plus an HTTPS front door.

Installation, end to end

```
# 1 · get the code and set the hospital's address
git clone https://github.com/abdulrazakucc/patient-nurse-schedule.git neostay
cd neostay && cp .env.example .env     # set NEOSTAY_TRUSTED_HOSTS and NEOSTAY_SITE_ADDRESS

# 2 · start it (with the bundled HTTPS proxy)
docker compose --profile https up -d --build

# 3 · create the first account (password typed at a hidden prompt)
docker compose exec app python -m app.accounts add first.admin@hospital.org --name "First Admin"

# 4 · check it end to end
NEOSTAY_SMOKE_PASSWORD='…' scripts/smoke_test.sh \
    https://neostay.hospital.local first.admin@hospital.org
```

TABLE 13 — MANAGING WHO CAN SIGN IN

Task	Command (run on the server)
Add a person	`docker compose exec app python -m app.accounts add name@hospital.org --name "Full Name"`
Reset a password	the same command again — it keeps the account and sets a new password
Remove a person	`docker compose exec app python -m app.accounts remove name@hospital.org`
List everyone	`docker compose exec app python -m app.accounts list`

Changes take effect immediately, with no restart. Removing an account or resetting a password signs that person out everywhere at once. The full guide, including single sign-on, backups and troubleshooting, is `docs/DEPLOYMENT.md`.

12 How correctness is checked

What runs before any change reaches a hospital or the published site.

TABLE 14 — THE AUTOMATED TEST SUITE (105 TESTS PASSING)

Test file	What it proves
`tests/test_engine_parity.py`	The browser engine and the Python engine give identical answers: 512 acuity classification cases plus a full unit schedule and 24-hour roster, compared value by value.
`tests/test_auth.py`	Passwords hash to the standard PBKDF2 vector; data and API refuse a signed-out request; removing an account or changing a password ends its sessions; repeated failures are slowed; production refuses unsafe settings.
`tests/test_pages_site.py`	The published site contains no readable data, email, name or password hash; every account opens the same data; a wrong password or removed account opens nothing; each build uses a new key; the browser's own decryption code opens a real build.
`tests/test_frontend_pages.py`	Every page loads the sign-in gate and nothing else; no page pulls data directly; no inline scripts.

Two workflows run on every push: the test suite, and a build of the real container which is then started and smoke-tested end to end with a throwaway account. The deployment workflow additionally refuses to publish if anything readable would be exposed.

> **Evidence behind this document.** Every figure quoted in Sections 5 to 9 was produced by executing the application's own code with the inputs shown, not transcribed by hand.

13 Adapting NeoStay to any hospital

Different units grade nurses differently. Here is what fits today, and what does not yet.

NeoStay currently models nurses on **Benner's four stages** and babies on the **N1–N4** levels of care. Hospitals use many other ladders: NHS Agenda for Change bands, clinical ladder tiers I–IV, or local titles such as "junior staff nurse" and "senior sister". Any of these can be used *today* by mapping local grades onto the four levels; deeper customisation is a code change, and is the first item on the roadmap in Section 14.

TABLE 15 — MAPPING LOCAL GRADES ONTO NEOSTAY'S FOUR LEVELS (WORKED EXAMPLES)

NeoStay level	NHS Agenda for Change	US clinical ladder	Typical local title	Counts as
L1 Novice	Band 5, first year	Clinical Nurse I	Newly qualified staff nurse	0.75
L2 Competent	Band 5, experienced	Clinical Nurse II	Staff nurse	0.90
L3 Proficient	Band 6	Clinical Nurse III	Senior staff nurse / charge	1.00
L4 Expert	Band 7+	Clinical Nurse IV	Sister / ANNP / unit lead	1.10

Practical advice for a unit adopting NeoStay this month. Agree the mapping once, with the nursing leadership, and write it down beside the roster. Enter the number of nurses in each of the four levels on the Scheduling page. Everything else — ratios, skill mix, rostering — follows automatically. Nothing in the application needs to change.

What is configurable today

Setting	Where	Changed by
Who may sign in	Accounts file on the server	NeoStay administrator, no restart
Nurses available per level	Scheduling page	Charge nurse, per session
Shifts per day (2 × 12 h or 3 × 8 h)	Scheduling page	Charge nurse, per session
Findings recorded per baby	Scheduling page	Charge nurse, per baby
Session length, host name, HTTPS, single sign-on	`.env`	Hospital IT

What is fixed in code today

Assumption	Current value	File to change
Number and names of nurse levels	4 (Benner stages)	`backend/app/nursing.py` and `frontend/engine.js`
Experience weighting per level	0.75 / 0.90 / 1.00 / 1.10	same two files
Novice cap and preceptor rule	30% of the bedside team; one preceptor per novice	same two files
Level of care → nurse level	N1→L1, N2→L2, N3→L3, N4→L4	`backend/app/acuity_tool.py`
Ratios per acuity column	1:1 / 1:2 / 1:2 / 1:3	`backend/app/acuity_tool.py`
Severity weights and phase shares	0.55 / 0.45; 0.15 + 0.45·S etc.	`backend/app/nursing.py`
The acuity tool itself	Abrazo Arrowhead tool as transcribed	`datasets/nurse-skills/*.csv`

Before any unit relies on this. The clinical lead should confirm the mapping from levels of care to nurse levels, the ratios used where the printed tool gives a range, and the local policy on novice supervision. These are planning assumptions, not regulations.

14 Future steps

What would make NeoStay usable by any hospital, in the order it should be built.

TABLE 16 — ROADMAP, HIGHEST VALUE FIRST

Step	Why it matters	Effort
1 · A staffing-policy file per hospital	Move nurse levels, their names, their weightings, the novice cap, the ratios and the level-of-care mapping out of the code into one reviewed configuration file, with a page in the application to edit it. A unit with five bands, or with 1:1 policies of its own, would then need no code change.	Medium
2 · Load the hospital's own acuity tool	The classifier already reads its rules from CSV files. Allowing a hospital to upload its own tool, validated on import, would let each unit classify by its own policy while keeping NeoStay's arithmetic.	Medium
3 · Use the hospital's own outcome data	Predictions currently rest on one centre's decade. An import step for a unit's own aggregated outcomes — same shape, validated — would make forecasts local, with the sample size shown per band.	Medium
4 · Named nurses and real rosters	Today the roster places anonymous nurses by level. Naming them, respecting leave, contracted hours, continuity of care and night-shift fairness, is what turns a plan into a publishable roster.	Large
5 · Save and reload a unit census	The census is lost when the tab closes. Storing it per unit — on the hospital server only — would allow shift handover, a record of what was planned, and later comparison with what happened.	Small
6 · Prospective validation	Compare forecast stay and forecast staffing with what actually happened, band by band, and publish the error. Without this, NeoStay stays a planning aid; with it, it can be cited.	Large
7 · Patient-level modelling	With patient-level data and ethical approval, survival and stay could be modelled directly (for example a competing-risks or survival model) instead of interpolated from bands, with uncertainty per baby.	Large
8 · Single sign-on everywhere, and an audit trail	Proxy-based single sign-on already exists; adding group-based roles (charge nurse, manager, read-only) and an access log would meet most hospital information-governance reviews.	Medium
9 · Multi-unit and multi-site view	One page comparing several units or hospitals, with per-unit thresholds, would support regional planning and transfers.	Medium
10 · Accessibility and offline use	A formal accessibility audit (WCAG 2.2 AA) and an offline-capable installable app would make NeoStay dependable on ward computers with poor connectivity.	Small

The through-line. Steps 1 to 3 remove every assumption that is specific to one hospital. After those three, adopting NeoStay becomes configuration rather than development — which is the condition for it to be useful to hospitals beyond the one it was written in.

15 Appendix — files, endpoints, settings

For the engineer who inherits this system.

Repository map

Path	Contents
`frontend/`	Seven pages, the sign-in gate, the engine, the design system, prebuilt data bundles.
`backend/app/`	FastAPI application, prediction, nursing model, acuity classifier, accounts and sign-in.
`datasets/`	Source outcome tables and the nurse-skills classifier (CSV, plus the original scan).
`scripts/`	Data-bundle build, published-site build, end-to-end smoke test.
`tests/`	Sign-in, published site, page gating, engine parity.
`deploy/`, `Dockerfile`, `docker-compose.yml`	Hospital deployment, including the optional HTTPS proxy.
`docs/`	User guide, deployment guide, and this document.

API endpoints

Method	Endpoint	Purpose
GET	`/api/health`	Service status (public, for monitoring).
POST	`/api/auth/login` · `/logout`	Sign in and out.
GET	`/api/auth/session`	Who is signed in (public).
GET	`/api/meta` · `/api/analytics`	Bands, nurse levels, aggregated distributions.
GET	`/api/acuity-tool`	The whole classifier, with the finding identifiers.
POST	`/api/predict`	Outcome and staffing forecast for one infant.
POST	`/api/schedule`	Unit staffing, skill mix and optional roster from a census.

Everything except `/api/health` and the sign-in routes requires a valid session.

Key settings

Variable	Meaning
`NEOSTAY_TRUSTED_HOSTS`	The host names the server answers to. Required on a server.
`NEOSTAY_AUTH_MODE`	`password` (default) or `proxy` for hospital single sign-on.
`NEOSTAY_SESSION_SECRET_FILE`	File holding the secret that signs session cookies; created on first start.
`NEOSTAY_SESSION_HOURS`	How long a sign-in lasts. Default 12.
`NEOSTAY_SITE_ADDRESS` · `NEOSTAY_TLS`	Address and certificate for the bundled HTTPS proxy.
`NEOSTAY_USERS_JSON`	Repository secret that seals the published GitHub Pages copy for a set of accounts.

NeoStay — Architecture and Methods. Project and Clinical Lead: Waseem Altaf, MD, Neonatologist. Technical Lead, Lead Developer and Applied AI/Data Science Lead: Abdul Razak, PhD. Decision support only — not for clinical use. © 2026 Waseem Altaf, MD. Licensed under the Apache License 2.0. Source document: `docs/architecture/neostay-architecture.html`.